

Assignment-5: Multi-Modal University Chatbot System

(CLO-3)

BSSE23 & BSCE22

Dr. Muhammad Asif

Human Computer Interaction

SE305T & MD445T (Spring-26)

Group Assessment

Information Technology University (ITU) – Lahore

06-May-2026

Total Marks = 100

1. Assignment Overview

This assignment asks your group to design and implement an interactive multi-modal chatbot system for a university environment. The system must allow users to interact using both text and voice input, and intelligently process queries related to university services.

The chatbot must support the following core functionalities:

- **Spam Detection:** identify and block irrelevant or malicious queries [20 marks]
- **Intent Identification:** classify user queries into predefined university-related intents [20 marks]
- **Response Generation:** generate meaningful, context-aware responses [20 marks]
- **Voice Interaction:** support speech-to-text (input) and text-to-speech (output) [20 marks]

Input Modes [20 marks]

Your system must allow the user to select input type at runtime:

1. Text Input Mode

- User types query via console or GUI
- Continuous interaction supported
- Suitable for testing classification and response

2. Voice Input Mode

- Capture audio using microphone
- Convert speech to text using STT
- Must display recognized text before processing

3. Hybrid Mode

- User can switch between text and voice dynamically
- System must handle both seamlessly

2. Background & Context

Modern universities receive thousands of repetitive queries daily regarding admissions, fees, courses, and schedules. Manual handling of these queries leads to:

- Increased workload on administrative staff
- Delayed responses to students
- Inefficient communication channels
- Exposure to spam or irrelevant queries

A multi-modal chatbot system can serve as an intelligent front-end interface, enabling:

- Automated query handling
- Voice-based accessibility
- Faster and consistent responses

Your task is to design such a system as a prototype solution for a university environment.

3. Required Libraries & Setup

Install the following Python libraries before starting:

Library	Install Command	Purpose
speech_recognition	<code>pip install SpeechRecognition</code>	Voice input (STT)
pyttsx3 / gTTS	<code>pip install pyttsx3</code>	Voice output (TTS)
sklearn / nltk	<code>pip install scikit-learn nltk</code>	Spam detection / intent classification
pyaudio (if needed)	<code>pip install pyaudio</code>	Microphone access
numpy	<code>pip install numpy</code>	Data handling

4. Technical Requirements

Your program must fulfill the following modules. Each module contains three graded tasks (T1, T2, T3).

4.1 Module 1: Spam Detection

Detect whether the user query is relevant to university context.

Task	Name	Description & Acceptance Criteria
T1	Basic Spam Classification	Classify input as Spam or Not Spam using rule-based or ML approach. Display result before further processing.
T2	Domain-Specific Filtering	Identify spam specific to university context (ads, irrelevant promotions). Must correctly classify at least 5 predefined spam cases.
T3	Spam Handling Response	If spam detected, system must block further processing and display: "This query is classified as spam."

4.2 Module 2: Intent Identification

Classify user queries into predefined intents.

Task	Name	Description & Acceptance Criteria
T1	Intent Classification	Identify at least 4 intents: Admission, Fee, Courses, Schedule. Display detected intent.
T2	Robust Mapping	Handle variations in queries (e.g., "fee", "charges", "semester cost"). Must correctly classify at least 10 test queries.
T3	Confidence Handling	Display confidence score or matching keyword pattern for detected intent.

4.3 Module 3: Response Generation

Generate meaningful responses based on detected intent.

Task	Name	Description & Acceptance Criteria
T1	Static Responses	Provide predefined responses for each intent.
T2	Dynamic Response Handling	Responses should adapt based on keywords (e.g., program-specific query).

T3	Error Handling	Handle unknown queries gracefully: “Sorry, I could not understand your request.”
-----------	----------------	--

4.4 Module 4: Voice Interaction

Enable full voice-based interaction.

Task	Name	Description & Acceptance Criteria
T1	Speech-to-Text	Convert voice input into text. Display recognized text.
T2	Text-to-Speech	Convert chatbot response into speech output.
T3	Integrated Voice Flow	Complete pipeline: Voice → Text → Processing → Voice Response. Must work in real time.

5. Code Structure Requirements

Your submitted code must follow these rules:

1. Single Python file: `hci_chatbot_YourGroupName.py`
2. Menu-driven system:
 - a. Select input mode (Text / Voice / Hybrid)
 - b. Run chatbot continuously
3. Modular implementation: Separate functions for each module
4. Each function must include Proper docstring
5. ESC key Exit chatbot
6. No global mutable variables
7. Code must run on a standard laptop

Submission Details:

- Deadline: Friday, May 15, 2026, at 11:59 PM.
- Python code
- PDF report
 - A table showing which member implemented which tasks in %.
 - Each member must write a short paragraph (min. 150 words) describing their personal contribution
- Demo Video (3-5 min)
- **Include: All group members name, student ID on the video.**
- Ensure clarity, coherence, and proper formatting in your responses.